

R&D Intensity and Firm Performance among European SMEs

A Panel Data Analysis

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1 Introduction

Research and development (R&D) investment is widely recognised as a key driver of firm competitiveness and long-run performance [1], [2]. Yet the short-run relationship between R&D spending and accounting-based performance measures such as return on assets (RoA) remains contested, particularly for small and medium-sized enterprises (SMEs) that face tighter resource constraints than large multinational firms [3], [4].

Under International Financial Reporting Standards (IFRS), R&D expenditures are generally expensed in the period they occur rather than capitalised as assets [5]. This accounting treatment creates a mechanical negative relationship between R&D intensity and current-period earnings: costs are recognised immediately while the economic benefits accrue with a lag of two to five years [2]. For SMEs, which often lack the financial slack to absorb these short-run earnings penalties, this dynamic may be particularly pronounced [6].

Building on the resource-based view [7], [8] and absorptive capacity theory [1], this research note examines two related questions for a sample of European EUR-reporting SMEs over 2015–2024. First, does R&D intensity negatively affect current-period RoA, consistent with the IFRS expensing effect? Second, does firm size moderate this relationship?

Research question: Does R&D intensity affect firm performance among European SMEs, and does firm size moderate this relationship?

2 Theoretical Background and Hypotheses

2.1 R&D Intensity and Firm Performance

The resource-based view holds that sustained competitive advantage derives from firm-specific resources that are valuable, rare, and inimitable [7], [8]. R&D investment generates precisely such resources — proprietary knowledge, patents, and technological capabilities that competitors cannot easily replicate [1]. In the long run, R&D intensity should therefore positively predict firm performance.

However, the accounting treatment of R&D under IFRS creates a short-run tension with this long-run expectation. R&D expenditures are charged against income in the period incurred, mechanically reducing net income and RoA [5]. The performance payoff materialises only after a multi-year lag [2], [9]. Empirical evidence for SMEs supports this pattern [6], [4].

H1: R&D intensity negatively affects current-period RoA among European SMEs, reflecting the immediate expensing effect under IFRS.

Test: $\beta(rd_intensity) < 0$ and statistically significant

2.2 The Moderating Role of Firm Size

The absorptive capacity framework [1] suggests that a firm’s ability to exploit internally generated knowledge depends on its organisational scale. Larger firms possess broader technical expertise, more developed commercialisation capabilities, and greater financial slack to sustain R&D programmes through the earnings-penalty phase [7], [8]. A larger SME can spread fixed R&D costs over a larger asset base, reducing the per-unit earnings impact [5].

H2: Firm size positively moderates the R&D intensity–RoA relationship — larger SMEs experience a smaller RoA penalty from R&D investment than smaller SMEs.

Test: $\beta(rd_intensity \times \ln_at) > 0$

3 Data and Method

3.1 Sample

The data are drawn from WRDS Compustat Global (comp_global_daily.g_funda), covering firms that report in EUR and meet the EU SME definition (≤ 250 employees or $\leq \text{€}43\text{m}$ total assets) over fiscal years 2015–2024. After applying data quality filters ($at > 0.1$, $sale > 0$, $seq > 0$, $at \geq 1$) and requiring at least three consecutive firm-year observations, the final sample comprises 8,116 firm-years from 1,233 unique firms across 33 European countries.

3.2 Variables

Dependent variable. Firm performance is measured as return on assets ($RoA = ib / at$), the standard accounting-based performance measure in SME research [3], [6].

Independent variable. R&D intensity is operationalised as R&D expenditure divided by total assets (xrd / at). Firms that do not separately report R&D are assigned zero, consistent with standard practice [5]. 35.8% of firm-years report positive R&D expenditure.

Moderator. Firm size is the natural logarithm of total assets ($\log(at)$) [7].

Controls. Leverage ($dltt / at$), capital expenditure intensity ($capx / at$), and cash ratio (che / at) control for capital structure, investment intensity, and liquidity [10]. All continuous variables except firm size are winsorized at the 1st–99th percentiles [11].

Variable	Field(s)	Formula	Role
RoA	ib, at	ib / at	Dependent (Y)
R&D intensity	xrd, at	$xrd.fillna(0) / at$	Independent (X)
R&D $\times$ Size	—	$rd_intensity \times \ln_at$	H2 interaction
Firm size	at	$\log(at)$	Moderator + Control

Variable	Field(s)	Formula	Role
Leverage	dltt, at	dltt / at	Control
CAPX intensity	capx, at	capx / at	Control
Cash ratio	che, at	che / at	Control

3.3 Estimation Strategy

We estimate two-way fixed effects (TWFE) panel regressions [12]:

$$\text{RoA}_{it} = \beta_1 \text{R\&D}_{it} + \beta_2 (\text{R\&D} \times \text{Size})_{it} + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it}$$

where μ_i are firm fixed effects (absorbing time-invariant characteristics including country, industry, and management quality), λ_t are year fixed effects (absorbing common macroeconomic shocks), and X_{it} is the vector of controls. Standard errors are clustered at the firm level [10]. The Hausman-style comparison between FE and RE estimates justifies fixed effects [13].

4 Results

4.1 Descriptive Statistics

Table 1: Descriptive statistics. Continuous variables winsorized at 1st–99th percentiles.

	N	mean	std	min	Median	max
RoA	8,116	−0.040000	0.194000	−0.955000	0.009000	0.354000
R&D Intensity	8,116	0.030000	0.077000	0.000000	0.000000	0.470000
Firm Size (log)	8,116	3.177000	1.300000	−2.254000	3.217000	9.669000
Leverage	8,116	0.137000	0.147000	0.000000	0.092000	0.638000
CAPX Intensity	8,116	0.034000	0.056000	0.000000	0.011000	0.315000
Cash Ratio	8,116	0.191000	0.196000	0.000000	0.126000	0.856000

The median RoA is 0.009, indicating that the typical firm is marginally profitable. The mean of −0.04 is pulled below the median by loss-making firms — a pattern typical of listed SME panels. R&D intensity has a median of zero: only 35.8% of firm-years report positive R&D expenditure, consistent with the low R&D reporting rate among European SMEs [4].

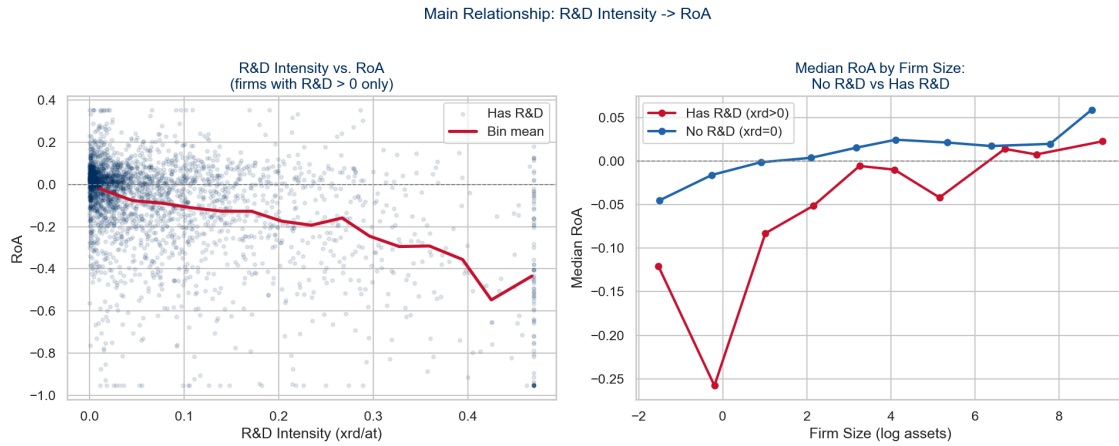


Figure 1: Main relationship: R&D intensity vs. RoA among R&D-active firms (left) and median RoA by firm size for firms with and without R&D (right).

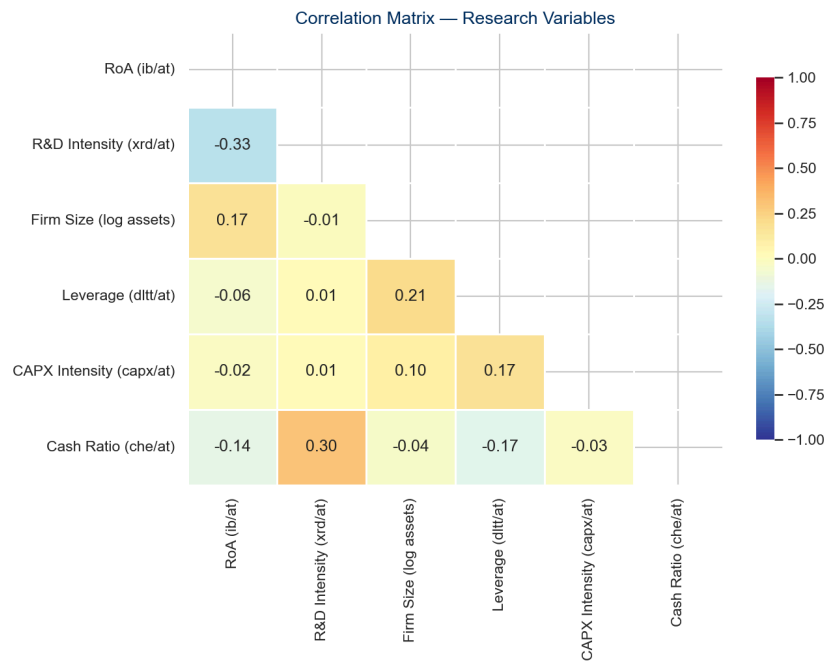


Figure 2: Correlation matrix of research variables.

4.2 Regression Results

Table 2: Panel regression results. Dependent variable: RoA. Standard errors in parentheses, clustered at firm level in Models (2)–(3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) OLS	(2) TWFE	(3) TWFE+H2
Variable			
rd_intensity	−0.7965***	−0.5346***	−0.6325***
rd_x_size	NaN	NaN	0.0371
ln_at	0.0277***	0.0466***	0.0459***
leverage	−0.1380***	−0.1336***	−0.1337***
capx_intensity	−0.0657	0.0187	0.0187
cash_ratio	−0.0531***	0.0313	0.0308
Firm FE	No	Yes	Yes
Year FE	No	Yes	Yes
Clustered SE	No	Yes	Yes
N	8,116	8,116	8,116
R ²	0.151	0.069	0.069

H1. R&D intensity is negative and statistically significant across all specifications. The TWFE estimate (Model 2) is $\beta = -0.535^{***}$ ($p < 0.01$), indicating that within the same firm, years with higher R&D intensity are associated with lower RoA, holding all other variables constant. **H1 is supported.** This is consistent with the IFRS expensing mechanism: R&D costs reduce current earnings while returns accrue with a multi-year lag [2], [5].

The OLS estimate (Model 1, $\beta = -0.796$) exceeds the TWFE estimate in magnitude by 33%, confirming omitted variable bias when firm fixed effects are omitted [13]. Unobserved time-invariant firm characteristics — such as innovation culture — that are positively correlated with R&D investment partially offset the negative earnings effect in the cross-section.

H2. The interaction between R&D intensity and firm size (Model 3) is positive ($\beta = 0.037$) but not statistically significant ($p = 0.559$). **H2 is not supported.** The direction is consistent with the absorptive capacity prediction [1], but the effect is too small to be distinguishable from zero in this sample.

5 Discussion and Conclusion

5.1 Discussion

The main finding — a negative and significant short-run effect of R&D intensity on RoA — is consistent with the IFRS expensing mechanism documented by Z. Griliches [2] and B. H. Hall and J. Lerner [5]. Firms that invest in R&D face an immediate earnings penalty that is only partially offset by future returns accruing beyond the observation window. The finding contributes to the SME innovation literature [4], [6] by documenting this short-run cost in a large, multi-country European panel. A policy implication follows: SMEs relying on accounting-based performance metrics may be disproportionately deterred from R&D investment [8].

5.2 Conclusion

This research note finds that R&D intensity significantly reduces current-period RoA among European SMEs ($\beta = -0.535$, $p < 0.01$), consistent with the IFRS expensing effect. Firm size does not significantly moderate this relationship. Several limitations apply: R&D activity is underestimated for non-reporting firms; the 10-year window may not capture the full lagged payoff; and Compustat Global coverage skews toward listed firms, limiting generalisability. Future research could exploit quasi-experimental variation in R&D tax incentives or extend the panel to capture delayed performance returns.

6 References

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